

STQM 270 Final Project

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STQM 270: Introduction to Data Mining

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Associations

Objectives

For the Associations model, we used the Associations.xlsx dataset to configure the model on the IBM SPSS Modeler. The dataset contains transaction data showing which types of books customers purchased together, and the goal is to identify meaningful relationships between these categories. We are trying to find patterns through association rule mining, which will reveal which items customers typically purchase together and the strength of their relationships. I applied three different association algorithms, which included Apriori, Carma, and Association Rules from IBM SPSS Modeler. The project achieves success when its results demonstrate dependable connections that businesses can use to make marketing and product placement decisions.

Description

The project uses Associations.xlsx as its dataset, which contains transactional records for multiple book categories, including CookBks, ChildBks, DoItYBks, ItalCook, ItalAtlas, and additional categories. The dataset contains customer transaction records, which show whether customers bought specific book types through each individual transaction. The dataset originally consisted of around 200 records, which the analysis later showed. We conducted multiple preprocessing activities before executing model testing. Since the data was originally in binary format (0 and 1), a reclassify node was applied to convert the values into more meaningful labels of yes and no, which made the results easier to interpret. All variables were treated as inputs because association rule mining does not require a target variable. These preprocessing decisions were made to improve clarity, accuracy, and usability of the final results.

Data Mining Session

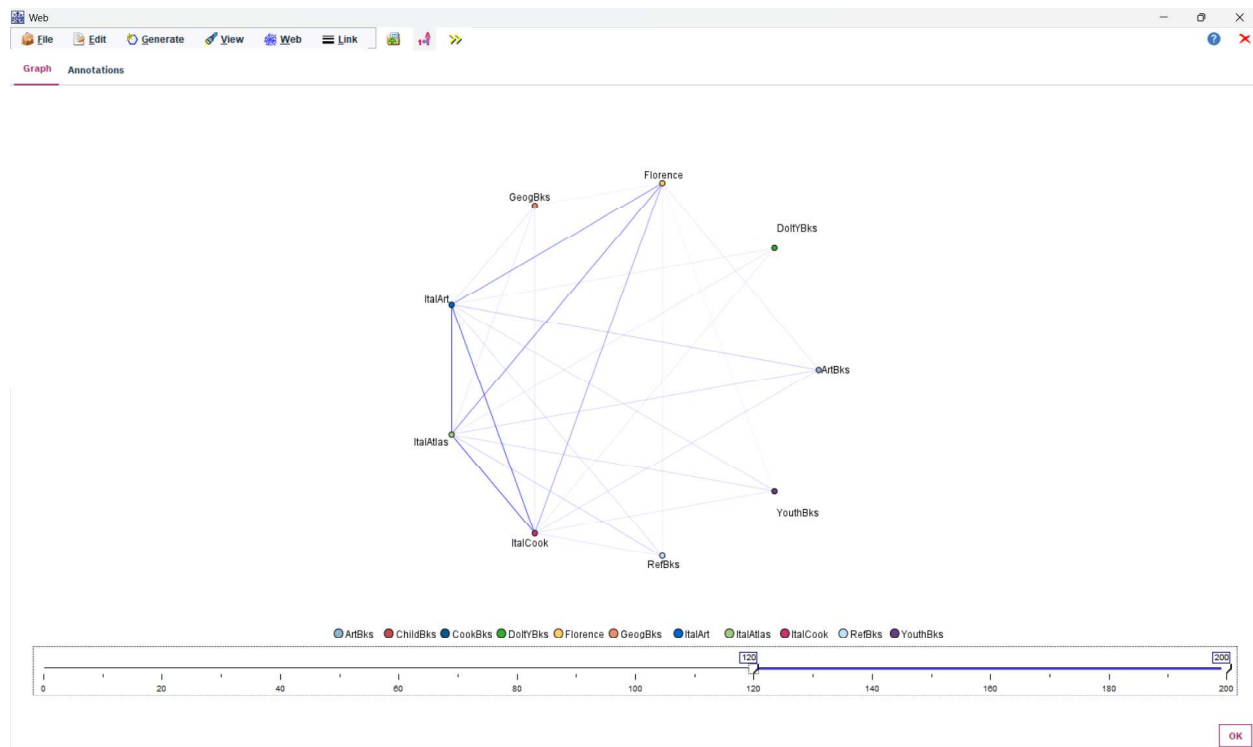
Preview from Associations.xlsx Node (11 fields, 10 records)

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Table Annotations

	ChildBks	YouthBks	CookBks	DoltYBks	RefBks	ArtBks	GeogBks	ItalCook	ItalAtlas	ItalArt	Florence
1	1.000	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	1.000	1.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000
3	0.000	0.000	1.000	1.000	0.000	0.000	1.000	1.000	0.000	0.000	0.000
4	0.000	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
6	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	0.000	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
10	1.000	1.000	1.000	1.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000

Above shows a preview of the association's dataset.



After importing our dataset, we connected a Web Graph node. Above shows a web graph of the Associations dataset. The thick lines on the graph represent a stronger relationship between two nodes in the dataset. We can see that ItalCook, ItalAtlas, and Florence are closely related, while categories like YouthBks and RefBks are weakly connected.

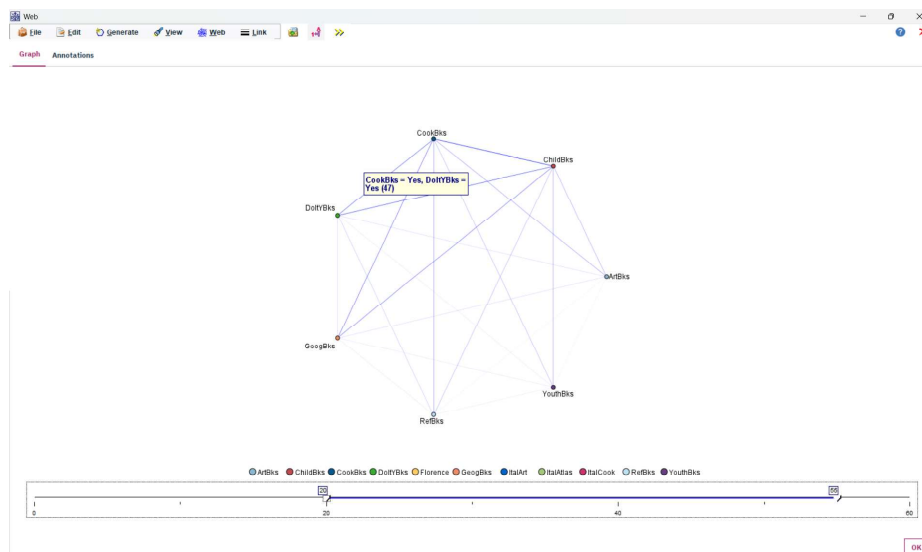
Data Audit of [11 fields]

File Edit Generate

Audit Quality Annotations

Field	Graph	Measurement	Min	Max	Mean	Std. Dev	Skewness	Unique	Valid
ChildBks		Flag	0.000	1.000	—	—	—	2	200
YouthBks		Flag	0.000	1.000	—	—	—	2	200
CookBks		Flag	0.000	1.000	—	—	—	2	200
DorYBks		Flag	0.000	1.000	—	—	—	2	200
RefBks		Flag	0.000	1.000	—	—	—	2	200
ArtBks		Flag	0.000	1.000	—	—	—	2	200
GeogBks		Flag	0.000	1.000	—	—	—	2	200
ItalCook		Flag	0.000	1.000	—	—	—	2	200
ItalAtlas		Flag	0.000	1.000	—	—	—	2	200
ItalArt		Flag	0.000	1.000	—	—	—	2	200
Florence		Flag	0.000	1.000	—	—	—	2	200

Next, we added a data audit node to help with data preparation. We can see in the graphs that it is taking the 0 as positive because it has a higher frequency. Under the Quality tab of this node, we can check for things like outliers (we are using a flag, so we don't have any outliers), what we should do if we have any missing values, and what action to take if we did have them, and more.



Our next goal was to change the 1 to a yes and the 0 to a no. To do this, we use the reclassify node. Next, we broke the original connection on our web graph node and connected it to the new reclassify node. When we hover our cursor over one of the lines, we can see 47 of the

records both have cookbooks and do-it-yourself books, meaning the customer bought both books at the same time, showing a correlation in the dataset.



Another option for visual representation of these correlations is through a directed web graph. In the image above, we see all of the possible correlations of purchases made when a child's book was bought, with the highest rule support being 54 children's books and children's books purchased together. This is possible for every category.

Apriori

Model Settings Summary Annotations

Sort by: Confidence %

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Consequent	Antecedent	Rule ID	Instances	Support %	Confidence	Rule Supp.	Lift	Deployability
CookBks	ArtBks	53	26	13.0	92.308	12.0	2.007	1.0
CookBks	DofyBks	12	22	11.0	90.909	10.0	1.976	1.0
CookBks	ChildBks	30	30	15.0	90.0	13.5	1.957	1.5
CookBks	GeogBks	34	30	15.0	90.0	13.5	1.957	1.5
ChildBks	ArtBks	54	27	13.5	88.889	12.0	1.932	1.5
ChildBks	DofyBks	15	26	13.0	88.462	11.5	1.923	1.5
CookBks	GeogBks	56	25	12.5	88.0	11.0	1.913	1.5
ChildBks	ArtBks	27	30	15.0	86.667	13.0	1.884	2.0
CookBks	GeogBks	28	30	15.0	86.667	13.0	1.884	2.0
ChildBks	ArtBks	29	30	15.0	86.667	13.0	1.884	2.0
ChildBks	DofyBks	44	21	10.5	85.714	9.0	1.863	1.5
ChildBks	GeogBks	13	27	13.5	85.185	11.5	1.852	2.0
CookBks	RefBks	47	26	13.0	84.615	11.0	1.839	2.0
CookBks	ArtBks	50	26	13.0	84.615	11.0	1.839	2.0

OK Cancel Apply Reset

Now we will move into modeling. We will start with the Apriori algorithm. After adding in the consequents and antecedents, we run the node. Taking a look at the results, we read the Consequent as the output and the Antecedent as the input. At the top, this means that if ArtBks, DoItYBks, and ChildBks, THEN CookBks. Additionally, we can get information like support %, confidence %, rule support %, lift, and deployability.

Consequent	Antecedent	Rule ID	Support %	Confidence %	Rule Support %	Lift	Deployability
CookBks	ItalCook	1	12.025	100.0	12.025	1.717	0.0
CookBks	DoItYBks	2	10.759	100.0	10.759	1.717	0.0
CookBks	ChildBks	3	9.494	100.0	9.494	1.717	0.0
CookBks	ChildBks DoItYBks	4	8.861	100.0	8.861	1.717	0.0
CookBks	ArtBks	5	7.595	100.0	7.595	1.717	0.0
DoItYBks	ArtBks	6	7.595	100.0	7.595	2.431	0.0
CookBks	ArtBks	7	7.595	100.0	7.595	3.362	0.0
CookBks	GeogBks	8	7.595	100.0	7.595	1.717	0.0
DoItYBks	GeogBks	9	7.595	100.0	7.595	1.717	0.0
CookBks	ArtBks	10	7.595	100.0	7.595	2.431	0.0
CookBks	DoItYBks	11	7.595	100.0	7.595	1.717	0.0
ArtBks	ItalCook	12	6.962	100.0	6.962	2.981	0.0
CookBks	DoItYBks	13	6.962	100.0	6.962	1.717	0.0
ArtBks	CookBks	14	6.329	100.0	6.329	2.981	0.0
CookBks	ChildBks	15	6.329	100.0	6.329	1.717	0.0

Next, we will use the Carma algorithm. After adjusting our fields and percentages for rule support and rule confidence, we run the node. We can see that we have some results that have 100% accuracy but 0% deployability. These are not good because if we deploy these rules, we are going to see and benefit.

Consequent	Antecedent	Rule ID	Support %	Confidence %	Rule Support %	Lift	Deployability
CoatBks	DoTfBks	348	41.139	72.308	29.747	1.242	11.392
ChildBks	DoTfBks	363	41.139	75.385	31.013	1.295	10.127
DoTfBks	ChildBks	349	34.177	72.222	24.684	1.756	9.494
ChildBks	GeogBks	304	37.975	75.0	28.481	1.288	9.494
ChildBks	ArBks	353	33.544	71.698	24.051	1.231	9.494
CoatBks	GeogBks	296	37.975	76.667	29.114	1.317	8.861
ChildBks	TrBks	324	34.177	74.074	25.316	1.272	8.861
CoatBks	ArBks	325	34.177	74.074	25.316	1.272	8.861
CoatBks	ArBks	302	33.544	75.472	25.316	1.280	8.228
ChildBks	CoatBks	326	29.114	73.913	21.519	1.269	7.595
CoatBks	ChildBks	301	28.481	75.556	21.519	1.288	6.962
ChildBks	CoatBks	347	25.316	72.5	18.354	1.245	6.962
CoatBks	ChildBks	278	31.013	79.592	24.684	1.367	6.329
DoTfBks	ChildBks	340	37.975	73.999	16.466	1.756	6.100
ChildBks	TrBks	351	22.785	73.222	16.456	1.24	6.329
ChildBks	CoatBks	287	25.316	77.5	19.62	1.331	5.696
GeogBks	ChildBks	358	19.62	70.968	13.924	1.869	5.696
ChildBks	CoatBks	227	29.747	82.979	24.684	1.425	5.063
ChildBks	DoTfBks	332	18.987	73.333	13.924	2.146	5.063
CoatBks	GeogBks	333	18.987	73.333	13.924	2.146	5.063

If we sort our results by deployability, we get more usable results. Now we can see what results have the highest values for deployability while still yielding good percentages in results like accuracy and confidence.

Rank	Rule ID	Condition	Prediction	Sorted By Deployability(%)	Condition Support (%)	Confidence (%)	Rule Support (%)	Lift
1	77			11.39	41.14	72.31	29.75	1.24
2	62			10.13	41.14	75.38	31.01	1.29
3	63			9.40	37.97	75.00	29.48	1.29
4	62			9.40	33.54	71.70	24.05	1.23
5	78			9.40	34.18	72.22	24.68	1.76
6	55			8.88	37.97	76.67	29.11	1.32
7	66			8.88	34.18	74.07	25.32	1.27
8	67			8.88	34.18	74.07	25.32	1.27
9	61			8.23	33.54	75.47	25.32	1.36
10	68			7.60	39.11	73.61	21.67	1.37
11	60			6.96	28.48	75.56	21.52	1.30
12	76			6.96	25.32	72.50	18.35	1.25
13	79			6.33	22.78	72.22	16.46	1.76
14	80			6.33	22.78	72.22	16.46	1.24
15	40			6.33	31.01	79.59	24.68	1.37
16	52			5.70	25.32	77.50	19.62	1.33
17	63			5.00	19.62	10.91	13.92	1.81
18	73			5.06	18.99	73.33	13.62	2.15
19	74			5.06	18.99	73.33	13.62	2.15
20	25			5.06	29.75	82.98	24.68	1.48

The next algorithm we will use is Association Rules. After adjusting our percentages to be the same as each of the other algorithms, we run the node. Now we have the results sorted by deployability. The only issue we have here is that we don't actually have the rule meaning if X then Y, if A then B, etc.

Association Rules

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Model Model Options Annotations

Output

- Model Information
 - Title
 - Build Settings Table
 - Excluded Inputs Table
- Interpretation
 - Title
 - Rules Table
 - Confidence
 - Condition Support
 - Rule Support
 - Deployability
 - Lift
 - Rules Word Cloud

Most Interesting Rules by Confidence

Rule 1- If ItalCook Then CookBks
 Rule 2- If DoltYBks & ItalCook Then CookBks
 Rule 3- If ChildBks & YouthBks & ArtBks Then CookBks
 Rule 4- If ChildBks & DoltYBks & ArtBks Then CookBks
 Rule 5- If YouthBks & ArtBks Then CookBks
 Rule 6- If DoltYBks & ArtBks Then CookBks
 Rule 7- If DoltYBks & GeogBks Then CookBks
 Rule 8- If ItalCook Then DoltYBks
 Rule 9- If ItalCook Then CookBks & DoltYBks
 Rule 10- If CookBks & ItalCook Then DoltYBks
 Rule 11- If DoltYBks & ArtBks & GeogBks Then ChildBks
 Rule 12- If DoltYBks & ArtBks & GeogBks Then CookBks
 Rule 13- If CookBks & DoltYBks & ArtBks Then ChildBks
 Rule 14- If YouthBks & DoltYBks Then ChildBks
 Rule 15- If ChildBks & DoltYBks & GeogBks Then CookBks
 Rule 16- If DoltYBks & ArtBks Then ChildBks
 Rule 17- If ArtBks & GeogBks Then ChildBks
 Rule 18- If ArtBks & GeogBks Then CookBks
 Rule 19- If YouthBks & CookBks & DoltYBks Then ChildBks
 Rule 20- If YouthBks & GeogBks Then ChildBks

OK Cancel Apply Reset

To fix the previous issue, we will go back into the association rules node and under the build options, we will check “create a sortable word cloud”. What this does is it actually gives us a layout of what the “if, then” rules are. This list is ordered by the highest yielding confidence.

Table (17 fields, 200 records)

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Table Annotations

	ChildBks	YouthBks	CookBks	DoltYBks	RelBks	ArtBks	GeogBks	ItalCook	ItalAtlas	ItalArt	Florence	SA-11 fields-1	SAC-11 fields-1	SA-Rule_ID-1	SA-11 fields-2	SAC-11 fields-2	SA-Rule_ID-2
1	Yes	Yes	Yes	No	No	No	No	No	No	No	No	DoltYBks	0.095	78	Smull6	Smull6	Smull6
2	Yes	Yes	No	No	No	Yes	No	No	No	No	No	ChildBks	0.095	82	Smull6	Smull6	Smull6
3	No	No	Yes	No	No	Yes	No	No	No	No	No	ChildBks	0.095	82	Smull6	Smull6	Smull6
4	No	No	No	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
5	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No	Yes	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
6	No	No	Yes	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
7	No	No	Yes	No	No	Yes	No	No	No	No	No	ChildBks	0.095	63	Smull6	Smull6	Smull6
8	Yes	Yes	Yes	No	No	No	Yes	No	No	No	Yes	ArtBks	0.092	83	CookBks & ArtBks	0.092	85
9	No	No	Yes	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
10	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73
11	Yes	No	No	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
12	Yes	No	Yes	No	Yes	No	No	No	No	No	No	DoltYBks	0.095	78	Smull6	Smull6	Smull6
13	No	No	No	No	Yes	No	No	No	No	No	Yes	ChildBks	0.095	82	CookBks	0.092	61
14	Yes	Yes	Yes	Yes	No	No	Yes	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
15	Yes	No	No	No	No	No	Yes	No	No	No	No	CookBks	0.089	55	Smull6	Smull6	Smull6
16	Yes	Yes	Yes	Yes	No	Yes	No	Yes	No	Yes	No	GeogBks	0.057	83	CookBks & GeogBks	0.025	65
17	No	No	No	Yes	No	No	No	No	No	No	No	CookBks	0.114	77	ChildBks	0.101	62
18	No	No	Yes	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
19	No	No	No	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
20	No	No	Yes	Yes	No	No	Yes	No	No	No	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73
21	Yes	No	Yes	Yes	Yes	Yes	Yes	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
22	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73
23	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	GeogBks	0.057	83	CookBks & GeogBks	0.025	65
24	Yes	No	Yes	Yes	Yes	Yes	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
25	No	No	No	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
26	No	No	Yes	Yes	No	No	Yes	Yes	No	No	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73
27	No	Yes	No	No	No	Yes	No	No	No	No	No	ChildBks	0.095	82	CookBks	0.082	61
28	Yes	No	No	Yes	No	No	No	No	No	No	No	CookBks	0.114	77	Smull6	Smull6	Smull6
29	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.070	76	Smull6	Smull6	Smull6
30	No	No	Yes	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
31	No	No	No	No	No	No	Yes	No	No	No	No	ChildBks	0.095	63	CookBks	0.089	55
32	Yes	No	Yes	No	No	No	No	No	No	No	No	DoltYBks	0.095	78	Smull6	Smull6	Smull6
33	No	Yes	No	No	No	No	No	No	No	No	No	ChildBks	0.089	66	Smull6	Smull6	Smull6
34	Yes	No	Yes	No	No	No	Yes	Yes	No	No	No	DoltYBks	0.095	78	ChildBks & Dolt...	0.032	70
35	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
36	No	No	No	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
37	Yes	No	No	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
38	Yes	No	Yes	Yes	Yes	Yes	No	Yes	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
39	Yes	No	Yes	Yes	No	Yes	No	Yes	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
40	Yes	No	No	No	Yes	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
41	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.070	76	Smull6	Smull6	Smull6
42	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.070	76	Smull6	Smull6	Smull6
43	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
44	Yes	No	Yes	No	Yes	Yes	Yes	No	No	Yes	Yes	DoltYBks	0.095	78	Smull6	Smull6	Smull6
45	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
46	No	No	No	No	No	No	No	No	No	No	No	Smull6	Smull6	Smull6	Smull6	Smull6	Smull6
47	No	No	No	No	No	No	No	No	No	No	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73

OK

If we then connect a table node to the results of our association rules algorithm, we get a representation of our data in addition to all other inputs. The issue here, as we can see above, is that if we have a null in one of the fields, all other categories will show up as null. In this case, we would like to focus on what the higher-yielding results are.

Table (17 fields, 97 records)

	ChildBks	YouthBks	CookBks	DoffBks	RefBks	ArtBks	GeogBks	ItalCook	ItalAtlas	ItalArt	Florence	SA-11 fields-1	SAC-11 fields-1	SA-Rule_ID-1	SA-11 fields-2	SAC-11 fields-2	SA-Rule_ID-2
1	Yes	Yes	Yes	No	No	No	No	No	No	No	No	DoItVBks	0.095	78	Small16	Small16	Small16
2	Yes	Yes	Yes	No	Yes	No	Yes	No	No	No	No	DoItVBks	0.095	78	Small16	Small16	Small16
3	No	No	Yes	No	No	Yes	No	No	No	No	No	ChildBks	0.095	82	Small16	Small16	Small16
4	No	No	Yes	No	No	No	Yes	No	No	No	No	ChildBks	0.095	63	Small16	Small16	Small16
5	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No	No	ArtBks	0.032	84	CookBks & ArtBks	0.032	85
6	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73
7	Yes	No	Yes	No	Yes	No	No	No	No	No	No	DoItVBks	0.095	78	Small16	Small16	Small16
9	No	No	No	No	No	Yes	No	No	No	No	Yes	ChildBks	0.095	82	CookBks	0.082	61
9	Yes	No	No	No	No	Yes	No	No	No	No	No	CookBks	0.089	55	Small16	Small16	Small16
10	Yes	Yes	Yes	No	Yes	No	Yes	No	Yes	No	Yes	GeogBks	0.057	83	CookBks & GeogBks	0.025	65
11	No	No	No	Yes	No	No	No	No	No	No	No	CookBks	0.114	77	ChildBks	0.101	62
12	No	No	Yes	Yes	No	No	No	No	No	No	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73
13	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73
14	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	GeogBks	0.057	83	CookBks & GeogBks	0.025	65
15	Yes	No	Yes	Yes	Yes	Yes	No	No	No	No	No	GeogBks	0.057	83	Small16	Small16	Small16
16	No	No	Yes	Yes	No	No	Yes	Yes	No	No	No	ChildBks	0.101	62	ChildBks & CookBks	0.051	73
17	No	Yes	No	No	No	Yes	No	No	No	No	No	ChildBks	0.095	62	CookBks	0.082	61
18	Yes	No	No	Yes	No	No	No	No	No	No	No	CookBks	0.114	77	Small16	Small16	Small16
19	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.070	76	Small16	Small16	Small16
20	No	No	No	No	No	No	Yes	No	No	No	No	ChildBks	0.095	63	CookBks	0.089	55
21	Yes	No	Yes	No	No	No	No	No	No	No	No	DoItVBks	0.095	78	Small16	Small16	Small16
22	No	Yes	No	No	No	No	No	No	No	No	No	ChildBks	0.089	66	Small16	Small16	Small16
23	Yes	No	Yes	No	No	No	Yes	Yes	No	No	No	DoItVBks	0.095	78	ChildBks & DoIt...	0.032	70
24	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.070	76	Small16	Small16	Small16
25	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.070	76	Small16	Small16	Small16
26	Yes	No	Yes	No	Yes	Yes	Yes	No	No	No	Yes	DoItVBks	0.095	78	Small16	Small16	Small16
27	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No	ChildBks	0.101	62	Small16	Small16	Small16
28	No	Yes	Yes	No	No	Yes	Yes	No	No	No	Yes	ChildBks	0.095	62	ChildBks & CookBks	0.051	73
29	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	No	DoItVBks	0.095	78	Small16	Small16	Small16
30	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	GeogBks	0.057	83	CookBks & GeogBks	0.025	65
31	No	No	No	Yes	Yes	No	No	No	No	No	No	CookBks	0.114	77	ChildBks	0.101	62
32	Yes	No	Yes	Yes	Yes	Yes	No	No	No	No	No	GeogBks	0.057	83	Small16	Small16	Small16
33	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	No	DoItVBks	0.095	78	Small16	Small16	Small16
34	No	No	No	Yes	No	No	No	No	No	No	No	CookBks	0.114	77	ChildBks	0.101	62
35	No	No	No	No	No	Yes	No	No	No	No	No	ChildBks	0.095	82	CookBks	0.082	61
36	Yes	No	Yes	Yes	Yes	Yes	No	No	Yes	No	No	GeogBks	0.057	83	Small16	Small16	Small16
37	Yes	Yes	No	Yes	Yes	No	No	No	No	No	No	CookBks	0.114	77	Small16	Small16	Small16
38	Yes	Yes	No	No	No	No	Yes	No	No	No	No	CookBks	0.089	55	Small16	Small16	Small16
39	No	No	No	No	No	Yes	No	No	No	No	No	ChildBks	0.095	82	CookBks	0.082	61
40	No	No	Yes	No	No	Yes	Yes	No	No	Yes	No	ChildBks	0.095	82	ChildBks & CookBks	0.051	73
41	No	Yes	Yes	No	No	No	No	No	No	No	No	ChildBks	0.089	66	Small16	Small16	Small16
42	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	ArtBks	0.032	84	CookBks & ArtBks	0.032	85
43	Yes	No	No	No	No	Yes	No	No	No	No	No	CookBks	0.082	61	Small16	Small16	Small16
44	No	No	No	No	Yes	No	No	No	No	No	No	CookBks	0.089	67	Small16	Small16	Small16
45	No	No	No	No	No	No	Yes	No	No	No	Yes	ChildBks	0.095	63	CookBks	0.089	55
46	No	No	Yes	No	No	No	Yes	No	No	No	No	ChildBks	0.095	63	Small16	Small16	Small16

Copy OK

To fix this, we attach a select node between our table and our results from the association rules algorithm. What this does is filter out all fields that have null values. Instead of the full 200 customers from before, we have a total of 97 customers.

Table (20 fields, 93 records)

File Edit Generate

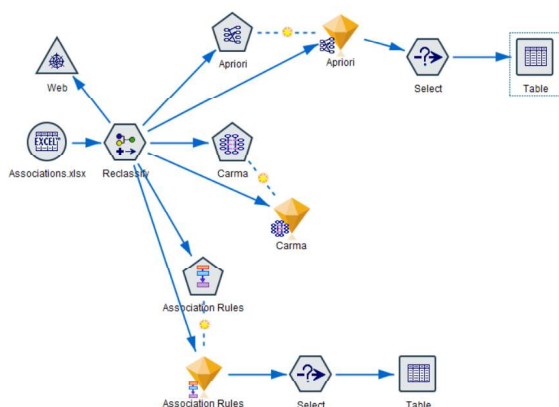
Table Annotations

	ChildBks	YouthBks	CookBks	DollYBks	RefBks	ArtBks	GeogBks	ItalCook	ItalAtlas	ItalArt	Florence	SA-11 fields-1	SAC-11 fields-1	SA-Rule_ID-1	SA-11 fields-2	SAC-11 fields-2	SA-Rule_ID-2	SA-11 fields-3	SA
1	Yes	Yes	No	No	No	No	No	No	No	No	No	DollYBks	0.722	39	5null16	5null16	5null16	5null16	
2	Yes	Yes	No	No	No	No	Yes	No	No	No	No	CookBks	0.778	14	5null16	5null16	5null16	5null16	
3	No	No	Yes	Yes	No	No	Yes	Yes	No	No	No	ChildBks	0.833	33	5null16	5null16	5null16	5null16	
4	No	No	No	Yes	No	No	No	No	No	No	No	ChildBks	0.754	7	CookBks	0.723	8	5null16	
5	No	No	No	Yes	No	No	No	No	No	No	No	ChildBks	0.754	7	CookBks	0.723	8	5null16	
6	No	No	No	No	Yes	No	No	No	No	No	No	CookBks	0.755	4	ChildBks	0.717	3	5null16	
7	Yes	Yes	No	Yes	No	Yes	No	No	No	No	No	DollYBks	0.755	49	5null16	5null16	5null16	5null16	
8	No	Yes	No	No	No	Yes	No	No	No	No	No	CookBks	0.909	12	ChildBks	0.741	1	5null16	
9	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	GeogBks	0.710	52	5null16	5null16	5null16	5null16	
10	Yes	No	Yes	No	No	No	Yes	No	No	No	No	DollYBks	0.723	36	5null16	5null16	5null16	5null16	
11	Yes	No	No	No	No	Yes	No	No	No	No	No	CookBks	0.767	6	5null16	5null16	5null16	5null16	
12	Yes	No	No	No	Yes	Yes	No	No	No	No	Yes	CookBks	0.867	28	DollYBks	0.722	23	5null16	
13	No	No	Yes	No	No	Yes	No	No	No	No	No	ChildBks	0.775	32	5null16	5null16	5null16	5null16	
14	Yes	Yes	No	No	No	Yes	No	No	No	No	No	CookBks	0.778	14	5null16	5null16	5null16	5null16	
15	Yes	No	No	Yes	No	No	No	No	No	No	No	CookBks	0.796	37	5null16	5null16	5null16	5null16	
16	No	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	ChildBks	0.889	54	5null16	5null16	5null16	5null16	
17	No	No	No	Yes	No	No	No	No	No	No	No	CookBks	0.741	2	5null16	5null16	5null16	5null16	
18	Yes	Yes	Yes	Yes	No	Yes	No	Yes	No	Yes	No	GeogBks	0.710	52	5null16	5null16	5null16	5null16	
19	No	No	Yes	No	No	No	Yes	No	No	No	Yes	ChildBks	0.750	5	5null16	5null16	5null16	5null16	
20	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.725	26	5null16	5null16	5null16	5null16	
21	Yes	No	Yes	Yes	No	Yes	No	No	No	No	No	GeogBks	0.710	52	5null16	5null16	5null16	5null16	
22	Yes	No	No	No	No	Yes	No	No	No	No	No	CookBks	0.816	31	5null16	5null16	5null16	5null16	
23	Yes	No	No	No	No	Yes	No	No	No	No	No	CookBks	0.816	31	5null16	5null16	5null16	5null16	
24	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.725	26	5null16	5null16	5null16	5null16	
25	Yes	Yes	No	No	Yes	Yes	Yes	No	No	No	No	CookBks	0.909	12	DollYBks	0.722	23	5null16	
26	No	No	No	Yes	Yes	No	No	No	No	No	Yes	ChildBks	0.900	30	ChildBks	0.867	29	5null16	
27	No	No	No	Yes	No	No	No	No	No	No	No	ChildBks	0.754	7	CookBks	0.723	8	5null16	
28	No	No	No	Yes	No	No	No	No	No	No	No	ChildBks	0.754	7	CookBks	0.723	8	5null16	
29	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	No	DollYBks	0.774	55	5null16	5null16	5null16	5null16	
30	No	No	Yes	No	Yes	No	No	No	No	No	No	ChildBks	0.725	26	5null16	5null16	5null16	5null16	
31	Yes	No	Yes	Yes	Yes	Yes	No	No	Yes	No	No	GeogBks	0.710	52	5null16	5null16	5null16	5null16	
32	No	No	Yes	Yes	Yes	No	Yes	No	No	No	No	ChildBks	0.833	33	5null16	5null16	5null16	5null16	
33	No	No	No	Yes	No	No	No	No	No	No	No	CookBks	0.741	2	5null16	5null16	5null16	5null16	
34	Yes	No	No	Yes	Yes	Yes	No	No	No	No	Yes	CookBks	0.923	53	5null16	5null16	5null16	5null16	
35	No	Yes	No	No	No	No	No	No	No	No	No	ChildBks	0.741	1	5null16	5null16	5null16	5null16	
36	Yes	No	No	No	No	No	Yes	No	No	No	No	CookBks	0.767	6	5null16	5null16	5null16	5null16	
37	No	No	No	No	No	No	Yes	No	No	No	No	CookBks	0.767	6	ChildBks	0.750	5	5null16	
38	No	No	No	Yes	Yes	No	No	No	No	No	No	CookBks	0.818	24	ChildBks	0.789	22	5null16	
39	No	No	No	Yes	No	No	No	No	No	No	No	ChildBks	0.754	7	CookBks	0.723	8	5null16	
40	Yes	Yes	No	Yes	Yes	No	No	No	No	No	No	CookBks	0.846	47	5null16	5null16	5null16	5null16	
41	Yes	No	Yes	No	Yes	Yes	Yes	No	No	No	Yes	DollYBks	0.774	66	5null16	5null16	5null16	5null16	
42	Yes	No	Yes	Yes	Yes	Yes	Yes	No	No	No	No	GeogBks	0.710	52	5null16	5null16	5null16	5null16	
43	No	No	Yes	No	No	No	Yes	No	No	No	No	ChildBks	0.750	5	5null16	5null16	5null16	5null16	
44	No	Yes	No	No	No	No	No	No	No	No	No	ChildBks	0.741	1	5null16	5null16	5null16	5null16	
45	Yes	No	Yes	No	No	No	No	No	No	No	No	DollYBks	0.722	39	5null16	5null16	5null16	5null16	
AR	Yes	No	Yes	No	No	No	No	No	No	No	No	CookBks	0.710	52	5null16	5null16	5null16	5null16	

OK

Next, if we copy and paste over our selection and table node to attach it to our Apriori algorithm, we can see the results as well. This table is sorted by the same category as our association rules algorithm because all we did was copy and paste the nodes. The best use of these tables is looking at which Rule_ID is related to the input. This means we can adjust our marketing strategy to sell more books to customers using the rules.

Results



Above is the final stream from our data mining session. The results of the association analysis revealed several strong relationships between different book categories. From the web graph visualization, it was clear that certain categories, such as ItalCook, ItalAtlas, and Florence, had strong connections, indicated by thicker lines representing stronger relationships. In contrast, categories like YouthBks and RefBks showed weaker relationships. Additional insights showed that specific combinations, such as CookBks and DoItYBks, appeared together frequently, with some relationships occurring in as many as 47 transactions. When applying the Apriori algorithm, rules such as if ArtBks, DoItYBks, and ChildBks then CookBks were identified, along with metrics like support, confidence, and lift that helped evaluate their strength. The Carma algorithm produced some rules with very high accuracy, but many had low deployability, meaning they were not useful in practice. By sorting results based on deployability, more meaningful and actionable rules were identified. The Association Rules algorithm further improved interpretability by clearly displaying if-then relationships and ranking them based on confidence. Overall, the results provided a strong understanding of customer purchasing patterns and highlighted the most valuable associations.

Recommendations

Based on the findings from the analysis, I would recommend that the client use these association rules to improve their marketing and sales strategies. Since certain book categories are frequently purchased together, these items should be placed near each other in stores or grouped in online recommendations. This can increase the likelihood of customers purchasing multiple items in a single transaction. Another recommendation is to create bundled promotions using the strongest associations identified in the analysis. For example, if customers frequently buy CookBks and DoItYBks together, offering a discount when both are purchased could

increase sales. Additionally, the client should focus on rules with high deployability rather than just high confidence, since those provide more practical value. These insights can also be used to improve recommendation systems and personalize customer experiences. By applying these strategies, the client can directly benefit from the patterns discovered in the data and make more informed business decisions.